a. What is the value of k in the decay equation for this isotope?

b. What is the isotope's mean life? (See Exercise 39.)

c. How long will it take 95% of a sample's radioactive nuclei to disintegrate?

41. Cooling soup Suppose that a cup of soup cooled from 90°C to 60°C after 10 min in a room whose temperature was 20°C. Use Newton's Law of Cooling to answer the following questions.

a. How much longer would it take the soup to cool to 35°C?

b. Instead of being left to stand in the room, the cup of 90°C soup is put in a freezer whose temperature is −15°C. How long will it take the soup to cool from 90°C to 35°C?

42. A beam of unknown temperature An aluminum beam was brought from the outside cold into a machine shop where the temperature was held at 65°F. After 10 min, the beam warmed to 35°F and after another 10 min it was 50°F. Use Newton's Law of Cooling to estimate the beam's initial temperature.

43. Surrounding medium of unknown temperature A pan of warm water (46°C) was put in a refrigerator. Ten minutes later, the water's temperature was 39°C; 10 min after that, it was 33°C. Use Newton's Law of Cooling to estimate how cold the refrigerator was.

44. Silver cooling in air The temperature of an ingot of silver is 60°C above room temperature right now. Twenty minutes ago, it was 70°C above room temperature. How far above room temperature will the silver be

a. 15 min from now?

b. 2 hours from now?

c. When will the silver be 10°C above room temperature?

45. The age of Crater Lake The charcoal from a tree killed in the volcanic eruption that formed Crater Lake in Oregon contained 44.5% of the carbon-14 found in living matter. About how old is Crater Lake?

46. The sensitivity of carbon-14 dating to measurement To see the effect of a relatively small error in the estimate of the amount of carbon-14 in a sample being dated, consider this hypothetical situation:

a. A bone fragment found in central Illinois in the year 2000 contains 17% of its original carbon-14 content. Estimate the year the animal died.

b. Repeat part (a), assuming 18% instead of 17%.

c. Repeat part (a), assuming 16% instead of 17%.

47. Carbon-14 The oldest known frozen human mummy, discovered in the Schnalstal glacier of the Italian Alps in 1991 and called *Otzi*, was found wearing straw shoes and a leather coat with goat fur, and holding a copper ax and stone dagger. It was estimated that Otzi died 5000 years before he was discovered in the melting glacier. How much of the original carbon-14 remained in Otzi at the time of his discovery?

48. Art forgery A painting attributed to Vermeer (1632–1675), which should contain no more than 96.2% of its original carbon-14, contains 99.5% instead. About how old is the forgery?

49. Lascaux Cave paintings Prehistoric cave paintings of animals were found in the Lascaux Cave in France in 1940. Scientific analysis revealed that only 15% of the original carbon-14 in the paintings remained. What is an estimate of the age of the paintings?

50. Incan mummy The frozen remains of a young Incan woman were discovered by archeologist Johan Reinhard on Mt. Ampato in Peru during an expedition in 1995.

a. How much of the original carbon-14 was present if the estimated age of the "Ice Maiden" was 500 years?

b. If a 1% error can occur in the carbon-14 measurement, what is the oldest possible age for the Ice Maiden?

7.5 Indeterminate Forms and L'Hôpital's Rule

HISTORICAL BIOGRAPHY
Guillaume François Antoine de l'Hôpital (1661–1704)
www.goo.gl/nMJIKA
Johann Bernoulli (1667–1748)
www.goo.gl/70BgHS

Expressions such as "$0/0$" and "∞/∞" look something like ordinary numbers. We say that they have the *form* of a number. But values cannot be assigned to them in a way that is consistent with the usual rules to add and multiply numbers. We call these expressions "indeterminate forms." Although they are not numbers, these indeterminate forms play a useful role in summarizing the limiting behavior of a function.

John (Johann) Bernoulli discovered a rule using derivatives to calculate limits of fractions whose numerators and denominators both approach zero or ∞. The rule is known today as **l'Hôpital's Rule**, after Guillaume de l'Hôpital, a French nobleman who wrote the earliest introductory differential calculus text, where the rule first appeared in print.

Indeterminate Form 0/0

If we want to know how the function

$$F(x) = \frac{3x - \sin x}{x}$$

behaves *near* $x = 0$ (where it is undefined), we can examine the limit of $F(x)$ as $x \to 0$. We cannot apply the Quotient Rule for limits (Theorem 1 of Chapter 2) because the limit of the denominator is 0. Moreover, in this case, *both* the numerator and denominator

approach 0, and $0/0$ is undefined. Such limits may or may not exist in general, but the limit does exist for the function $F(x)$ under discussion by applying l'Hôpital's Rule, as we will see in Example 1d.

If the continuous functions $f(x)$ and $g(x)$ are both zero at $x = a$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)}$$

cannot be found by substituting $x = a$. The substitution produces $0/0$, a meaningless expression, which we cannot evaluate. We use $0/0$ as a notation for an expression that does not have a numerical value, known as an **indeterminate form**. Other meaningless expressions often occur, such as ∞/∞, $\infty \cdot 0$, $\infty - \infty$, 0^0, and 1^∞, which cannot be evaluated in a consistent way; these are called indeterminate forms as well. Sometimes, but not always, limits that lead to indeterminate forms may be found by cancelation, rearrangement of terms, or other algebraic manipulations. This was our experience in Chapter 2. It took considerable analysis in Section 2.4 to find $\lim_{x \to 0} (\sin x)/x$. But we have had success with the limit

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a},$$

from which we calculate derivatives and which produces the indeterminant form $0/0$ if we attempt to substitute $x = a$. L'Hôpital's Rule enables us to draw on our success with derivatives to evaluate limits for which substitution leads to indeterminate forms.

THEOREM 5—L'Hôpital's Rule

Suppose that $f(a) = g(a) = 0$, that f and g are differentiable on an open interval I containing a, and that $g'(x) \neq 0$ on I if $x \neq a$. Then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

assuming that the limit on the right side of this equation exists.

We give a proof of Theorem 5 at the end of this section.

Caution

To apply l'Hôpital's Rule to f/g, divide the derivative of f by the derivative of g. Do not fall into the trap of taking the derivative of f/g. The quotient to use is f'/g', not $(f/g)'$.

EXAMPLE 1 The following limits involve $0/0$ indeterminate forms, so we apply l'Hôpital's Rule. In some cases, it must be applied repeatedly.

(a) $$\lim_{x \to 0} \frac{3x - \sin x}{x} = \lim_{x \to 0} \frac{3 - \cos x}{1} = \left. \frac{3 - \cos x}{1} \right|_{x=0} = 2$$

(b) $$\lim_{x \to 0} \frac{\sqrt{1 + x} - 1}{x} = \lim_{x \to 0} \frac{\frac{1}{2\sqrt{1 + x}}}{1} = \frac{1}{2}$$

(c) $$\lim_{x \to 0} \frac{\sqrt{1 + x} - 1 - x/2}{x^2}$$ $\frac{0}{0}$; apply l'Hôpital's Rule.

$$= \lim_{x \to 0} \frac{(1/2)(1 + x)^{-1/2} - 1/2}{2x}$$ Still $\frac{0}{0}$; apply l'Hôpital's Rule again.

$$= \lim_{x \to 0} \frac{-(1/4)(1 + x)^{-3/2}}{2} = -\frac{1}{8}$$ Not $\frac{0}{0}$; limit is found.

(d) $\lim_{x\to 0} \dfrac{x - \sin x}{x^3}$ $\quad \frac{0}{0}$; apply l'Hôpital's Rule.

$= \lim_{x\to 0} \dfrac{1 - \cos x}{3x^2}$ Still $\frac{0}{0}$; apply l'Hôpital's Rule again.

$= \lim_{x\to 0} \dfrac{\sin x}{6x}$ Still $\frac{0}{0}$; apply l'Hôpital's Rule again.

$= \lim_{x\to 0} \dfrac{\cos x}{6} = \dfrac{1}{6}$ Not $\frac{0}{0}$; limit is found. ■

Here is a summary of the procedure we followed in Example 1.

Using L'Hôpital's Rule

To find

$$\lim_{x\to a} \frac{f(x)}{g(x)}$$

by l'Hôpital's Rule, we continue to differentiate f and g, so long as we still get the form $0/0$ at $x = a$. But as soon as one or the other of these derivatives is different from zero at $x = a$ we stop differentiating. L'Hôpital's Rule does not apply when either the numerator or denominator has a finite nonzero limit.

EXAMPLE 2 Be careful to apply l'Hôpital's Rule correctly:

$\lim_{x\to 0} \dfrac{1 - \cos x}{x + x^2}$ $\quad \frac{0}{0}$

$= \lim_{x\to 0} \dfrac{\sin x}{1 + 2x}$ Not $\frac{0}{0}$

It is tempting to try to apply l'Hôpital's Rule again, which would result in

$$\lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2},$$

but this is not the correct limit. L'Hôpital's Rule can be applied only to limits that give indeterminate forms, and $\lim_{x\to 0}(\sin x)/(1 + 2x)$ does not give an indeterminate form. Instead, this limit is $0/1 = 0$, and the correct answer for the original limit is 0. ■

L'Hôpital's Rule applies to one-sided limits as well.

EXAMPLE 3 In this example the one-sided limits are different.

(a) $\lim_{x\to 0^+} \dfrac{\sin x}{x^2}$ $\quad \frac{0}{0}$

$= \lim_{x\to 0^+} \dfrac{\cos x}{2x} = \infty$ Positive for $x > 0$

(b) $\lim_{x\to 0^-} \dfrac{\sin x}{x^2}$ $\quad \frac{0}{0}$

$= \lim_{x\to 0^-} \dfrac{\cos x}{2x} = -\infty$ Negative for $x < 0$ ■

Recall that ∞ and $+\infty$ mean the same thing.

Indeterminate Forms ∞/∞, $\infty \cdot 0$, $\infty - \infty$

Sometimes when we try to evaluate a limit as $x \to a$ by substituting $x = a$ we get an indeterminant form like ∞/∞, $\infty \cdot 0$, or $\infty - \infty$, instead of $0/0$. We first consider the form ∞/∞.

More advanced treatments of calculus prove that l'Hôpital's Rule applies to the indeterminate form ∞/∞, as well as to $0/0$. If $f(x) \to \pm\infty$ and $g(x) \to \pm\infty$ as $x \to a$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists. In the notation $x \to a$, a may be either finite or infinite. Moreover, $x \to a$ may be replaced by the one-sided limits $x \to a^+$ or $x \to a^-$.

EXAMPLE 4 Find the limits of these ∞/∞ forms:

(a) $\lim_{x\to\pi/2} \dfrac{\sec x}{1 + \tan x}$ **(b)** $\lim_{x\to\infty} \dfrac{\ln x}{2\sqrt{x}}$ **(c)** $\lim_{x\to\infty} \dfrac{e^x}{x^2}$.

Solution

(a) The numerator and denominator are discontinuous at $x = \pi/2$, so we investigate the one-sided limits there. To apply l'Hôpital's Rule, we can choose I to be any open interval with $x = \pi/2$ as an endpoint.

$$\lim_{x\to(\pi/2)^-} \frac{\sec x}{1 + \tan x} \qquad \frac{\infty}{\infty} \text{ from the left so we apply l'Hôpital's Rule.}$$

$$= \lim_{x\to(\pi/2)^-} \frac{\sec x \tan x}{\sec^2 x} = \lim_{x\to(\pi/2)^-} \sin x = 1$$

The right-hand limit is 1 also, with $(-\infty)/(-\infty)$ as the indeterminate form. Therefore, the two-sided limit is equal to 1.

(b) $\lim_{x\to\infty} \dfrac{\ln x}{2\sqrt{x}} = \lim_{x\to\infty} \dfrac{1/x}{1/\sqrt{x}} = \lim_{x\to\infty} \dfrac{1}{\sqrt{x}} = 0$ $\qquad \dfrac{1/x}{1/\sqrt{x}} = \dfrac{\sqrt{x}}{x} = \dfrac{1}{\sqrt{x}}$

(c) $\lim_{x\to\infty} \dfrac{e^x}{x^2} = \lim_{x\to\infty} \dfrac{e^x}{2x} = \lim_{x\to\infty} \dfrac{e^x}{2} = \infty$ ■

Next we turn our attention to the indeterminate forms $\infty \cdot 0$ and $\infty - \infty$. Sometimes these forms can be handled by using algebra to convert them to a $0/0$ or ∞/∞ form. Here again we do not mean to suggest that $\infty \cdot 0$ or $\infty - \infty$ is a number. They are only notations for functional behaviors when considering limits. Here are examples of how we might work with these indeterminate forms.

EXAMPLE 5 Find the limits of these $\infty \cdot 0$ forms:

(a) $\lim_{x\to\infty} \left(x \sin \dfrac{1}{x}\right)$ **(b)** $\lim_{x\to 0^+} \sqrt{x} \ln x$

Solution

(a) $\lim_{x\to\infty} \left(x \sin \dfrac{1}{x}\right) = \lim_{h\to 0^+} \left(\dfrac{1}{h} \sin h\right) = \lim_{h\to 0^+} \dfrac{\sin h}{h} = 1$ $\qquad \infty \cdot 0$; let $h = 1/x$.

(b) $\lim_{x\to 0^+} \sqrt{x} \ln x = \lim_{x\to 0^+} \dfrac{\ln x}{1/\sqrt{x}}$ $\qquad \infty \cdot 0$ converted to ∞/∞

$$= \lim_{x\to 0^+} \frac{1/x}{-1/2x^{3/2}} \qquad \text{l'Hôpital's Rule applied}$$

$$= \lim_{x\to 0^+} \left(-2\sqrt{x}\right) = 0$$ ■

EXAMPLE 6 Find the limit of this $\infty - \infty$ form:

$$\lim_{x\to 0} \left(\frac{1}{\sin x} - \frac{1}{x}\right).$$

Solution If $x \to 0^+$, then $\sin x \to 0^+$ and

$$\frac{1}{\sin x} - \frac{1}{x} \to \infty - \infty.$$

Similarly, if $x \to 0^-$, then $\sin x \to 0^-$ and

$$\frac{1}{\sin x} - \frac{1}{x} \to -\infty - (-\infty) = -\infty + \infty.$$

Neither form reveals what happens in the limit. To find out, we first combine the fractions:

$$\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x}. \qquad \text{Common denominator is } x \sin x.$$

Then we apply l'Hôpital's Rule to the result:

$$\lim_{x\to 0}\left(\frac{1}{\sin x} - \frac{1}{x}\right) = \lim_{x\to 0} \frac{x - \sin x}{x \sin x} \qquad \frac{0}{0}$$

$$= \lim_{x\to 0} \frac{1 - \cos x}{\sin x + x \cos x} \qquad \text{Still } \frac{0}{0}$$

$$= \lim_{x\to 0} \frac{\sin x}{2\cos x - x \sin x} = \frac{0}{2} = 0.$$

Indeterminate Powers

Limits that lead to the indeterminate forms 1^∞, 0^0, and ∞^0 can sometimes be handled by first taking the logarithm of the function. We use l'Hôpital's Rule to find the limit of the logarithm expression and then exponentiate the result to find the original function limit. This procedure is justified by the continuity of the exponential function and Theorem 10 in Section 2.5, and it is formulated as follows. (The formula is also valid for one-sided limits.)

If $\lim_{x\to a} \ln f(x) = L$, then

$$\lim_{x\to a} f(x) = \lim_{x\to a} e^{\ln f(x)} = e^L.$$

Here a may be either finite or infinite.

EXAMPLE 7 Apply l'Hôpital's Rule to show that $\lim_{x\to 0^+}(1 + x)^{1/x} = e$.

Solution The limit leads to the indeterminate form 1^∞. We let $f(x) = (1 + x)^{1/x}$ and find $\lim_{x\to 0^+} \ln f(x)$. Since

$$\ln f(x) = \ln (1 + x)^{1/x} = \frac{1}{x}\ln (1 + x),$$

l'Hôpital's Rule now applies to give

$$\lim_{x\to 0^+} \ln f(x) = \lim_{x\to 0^+} \frac{\ln (1 + x)}{x} \qquad \frac{0}{0}$$

$$= \lim_{x\to 0^+} \frac{\frac{1}{1 + x}}{1} \qquad \text{l'Hôpital's Rule applied}$$

$$= \frac{1}{1} = 1.$$

Therefore, $\lim_{x\to 0^+} (1 + x)^{1/x} = \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} e^{\ln f(x)} = e^1 = e.$

EXAMPLE 8 Find $\lim_{x\to\infty} x^{1/x}$.

Solution The limit leads to the indeterminate form ∞^0. We let $f(x) = x^{1/x}$ and find $\lim_{x\to\infty} \ln f(x)$. Since

$$\ln f(x) = \ln x^{1/x} = \frac{\ln x}{x},$$

l'Hôpital's Rule gives

$$\lim_{x\to\infty} \ln f(x) = \lim_{x\to\infty} \frac{\ln x}{x} \qquad \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty} \frac{1/x}{1} \qquad \text{l'Hôpital's Rule applied}$$

$$= \frac{0}{1} = 0.$$

Therefore $\lim_{x\to\infty} x^{1/x} = \lim_{x\to\infty} f(x) = \lim_{x\to\infty} e^{\ln f(x)} = e^0 = 1.$ ■

Proof of L'Hôpital's Rule

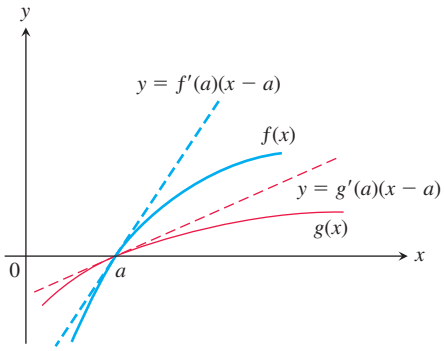


FIGURE 7.19 The two functions in l'Hôpital's Rule, graphed with their linear approximations at $x = a$.

HISTORICAL BIOGRAPHY
Augustin-Louis Cauchy (1789–1857)
www.goo.gl/ORQbxf

Before we prove l'Hôpital's Rule, we consider a special case to provide some geometric insight for its reasonableness. Consider the two functions $f(x)$ and $g(x)$ having *continuous* derivatives and satisfying $f(a) = g(a) = 0$, $g'(a) \neq 0$. The graphs of $f(x)$ and $g(x)$, together with their linearizations $y = f'(a)(x - a)$ and $y = g'(a)(x - a)$, are shown in Figure 7.19. We know that near $x = a$, the linearizations provide good approximations to the functions. In fact,

$$f(x) = f'(a)(x - a) + \varepsilon_1(x - a) \quad \text{and} \quad g(x) = g'(a)(x - a) + \varepsilon_2(x - a)$$

where $\varepsilon_1 \to 0$ and $\varepsilon_2 \to 0$ as $x \to a$. So, as Figure 7.19 suggests,

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(a)(x - a) + \varepsilon_1(x - a)}{g'(a)(x - a) + \varepsilon_2(x - a)}$$

$$= \lim_{x\to a} \frac{f'(a) + \varepsilon_1}{g'(a) + \varepsilon_2} = \frac{f'(a)}{g'(a)} \qquad g'(a) \neq 0$$

$$= \lim_{x\to a} \frac{f'(x)}{g'(x)}, \qquad \text{Continuous derivatives}$$

as asserted by l'Hôpital's Rule. We now proceed to a proof of the rule based on the more general assumptions stated in Theorem 5, which do not require that $g'(a) \neq 0$ and that the two functions have *continuous* derivatives.

The proof of l'Hôpital's Rule is based on Cauchy's Mean Value Theorem, an extension of the Mean Value Theorem that involves two functions instead of one. We prove Cauchy's Theorem first and then show how it leads to l'Hôpital's Rule.

When $g(x) = x$, Theorem 6 is the Mean Value Theorem in Chapter 4.

THEOREM 6—Cauchy's Mean Value Theorem
Suppose functions f and g are continuous on $[a, b]$ and differentiable throughout (a, b) and also suppose $g'(x) \neq 0$ throughout (a, b). Then there exists a number c in (a, b) at which

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Proof We apply the Mean Value Theorem of Section 4.2 twice. First we use it to show that $g(a) \neq g(b)$. For if $g(b)$ did equal $g(a)$, then the Mean Value Theorem would give

$$g'(c) = \frac{g(b) - g(a)}{b - a} = 0$$

for some c between a and b, which cannot happen because $g'(x) \neq 0$ in (a, b).

We next apply the Mean Value Theorem to the function

$$F(x) = f(x) - f(a) - \frac{f(b) - f(a)}{g(b) - g(a)}[\,g(x) - g(a)\,].$$

This function is continuous and differentiable where f and g are, and $F(b) = F(a) = 0$. Therefore, there is a number c between a and b for which $F'(c) = 0$. When expressed in terms of f and g, this equation becomes

$$F'(c) = f'(c) - \frac{f(b) - f(a)}{g(b) - g(a)}[\,g'(c)\,] = 0$$

so that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

■

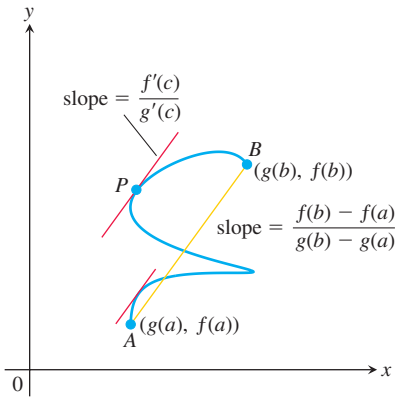


FIGURE 7.20 There is at least one point P on the curve C for which the slope of the tangent line to the curve at P is the same as the slope of the secant line joining the points $A(g(a), f(a))$ and $B(g(b), f(b))$.

Cauchy's Mean Value Theorem has a geometric interpretation for a general winding curve C in the plane joining the two points $A = (g(a), f(a))$ and $B = (g(b), f(b))$. In Chapter 11 you will learn how the curve C can be formulated to show that there is at least one point P on the curve for which the tangent line to the curve at P is parallel to the secant line joining the points A and B. The slope of that tangent line turns out to be the quotient f'/g' evaluated at the number c in the interval (a, b), which is the left-hand side of the equation in Theorem 6. Because the slope of the secant line joining A and B is

$$\frac{f(b) - f(a)}{g(b) - g(a)},$$

the equation in Cauchy's Mean Value Theorem says that the slope of the tangent line equals the slope of the secant line. This geometric interpretation is shown in Figure 7.20. Notice from the figure that it is possible for more than one point on the curve C to have a tangent line that is parallel to the secant line joining A and B.

Proof of l'Hôpital's Rule We first establish the limit equation for the case $x \to a^+$. The method needs almost no change to apply to $x \to a^-$, and the combination of these two cases establishes the result.

Suppose that x lies to the right of a. Then $g'(x) \neq 0$, and we can apply Cauchy's Mean Value Theorem to the closed interval from a to x. This step produces a number c between a and x such that

$$\frac{f'(c)}{g'(c)} = \frac{f(x) - f(a)}{g(x) - g(a)}.$$

But $f(a) = g(a) = 0$, so

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)}.$$

As x approaches a, c approaches a because it always lies between a and x. Therefore,

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = \lim_{c \to a^+} \frac{f'(c)}{g'(c)} = \lim_{x \to a^+} \frac{f'(x)}{g'(x)},$$

which establishes l'Hôpital's Rule for the case where x approaches a from above. The case where x approaches a from below is proved by applying Cauchy's Mean Value Theorem to the closed interval $[\,x, a\,]$, $x < a$. ■

EXERCISES 7.5

Finding Limits in Two Ways

In Exercises 1–6, use l'Hôpital's Rule to evaluate the limit. Then evaluate the limit using a method studied in Chapter 2.

1. $\lim_{x\to -2} \dfrac{x+2}{x^2-4}$
2. $\lim_{x\to 0} \dfrac{\sin 5x}{x}$
3. $\lim_{x\to \infty} \dfrac{5x^2-3x}{7x^2+1}$
4. $\lim_{x\to 1} \dfrac{x^3-1}{4x^3-x-3}$
5. $\lim_{x\to 0} \dfrac{1-\cos x}{x^2}$
6. $\lim_{x\to \infty} \dfrac{2x^2+3x}{x^3+x+1}$

Applying l'Hôpital's Rule

Use l'Hôpital's rule to find the limits in Exercises 7–50.

7. $\lim_{x\to 2} \dfrac{x-2}{x^2-4}$
8. $\lim_{x\to -5} \dfrac{x^2-25}{x+5}$
9. $\lim_{t\to -3} \dfrac{t^3-4t+15}{t^2-t-12}$
10. $\lim_{t\to -1} \dfrac{3t^3+3}{4t^3-t+3}$
11. $\lim_{x\to \infty} \dfrac{5x^3-2x}{7x^3+3}$
12. $\lim_{x\to \infty} \dfrac{x-8x^2}{12x^2+5x}$
13. $\lim_{t\to 0} \dfrac{\sin t^2}{t}$
14. $\lim_{t\to 0} \dfrac{\sin 5t}{2t}$
15. $\lim_{x\to 0} \dfrac{8x^2}{\cos x-1}$
16. $\lim_{x\to 0} \dfrac{\sin x-x}{x^3}$
17. $\lim_{\theta\to \pi/2} \dfrac{2\theta-\pi}{\cos(2\pi-\theta)}$
18. $\lim_{\theta\to -\pi/3} \dfrac{3\theta+\pi}{\sin(\theta+(\pi/3))}$
19. $\lim_{\theta\to \pi/2} \dfrac{1-\sin\theta}{1+\cos 2\theta}$
20. $\lim_{x\to 1} \dfrac{x-1}{\ln x-\sin \pi x}$
21. $\lim_{x\to 0} \dfrac{x^2}{\ln(\sec x)}$
22. $\lim_{x\to \pi/2} \dfrac{\ln(\csc x)}{(x-(\pi/2))^2}$
23. $\lim_{t\to 0} \dfrac{t(1-\cos t)}{t-\sin t}$
24. $\lim_{t\to 0} \dfrac{t\sin t}{1-\cos t}$
25. $\lim_{x\to (\pi/2)^-} \left(x-\dfrac{\pi}{2}\right)\sec x$
26. $\lim_{x\to (\pi/2)^-} \left(\dfrac{\pi}{2}-x\right)\tan x$
27. $\lim_{\theta\to 0} \dfrac{3^{\sin\theta}-1}{\theta}$
28. $\lim_{\theta\to 0} \dfrac{(1/2)^\theta-1}{\theta}$
29. $\lim_{x\to 0} \dfrac{x2^x}{2^x-1}$
30. $\lim_{x\to 0} \dfrac{3^x-1}{2^x-1}$
31. $\lim_{x\to \infty} \dfrac{\ln(x+1)}{\log_2 x}$
32. $\lim_{x\to \infty} \dfrac{\log_2 x}{\log_3(x+3)}$
33. $\lim_{x\to 0^+} \dfrac{\ln(x^2+2x)}{\ln x}$
34. $\lim_{x\to 0^+} \dfrac{\ln(e^x-1)}{\ln x}$
35. $\lim_{y\to 0} \dfrac{\sqrt{5y+25}-5}{y}$
36. $\lim_{y\to 0} \dfrac{\sqrt{ay+a^2}-a}{y}, \quad a>0$
37. $\lim_{x\to \infty} (\ln 2x-\ln(x+1))$
38. $\lim_{x\to 0^+} (\ln x-\ln\sin x)$
39. $\lim_{x\to 0^+} \dfrac{(\ln x)^2}{\ln(\sin x)}$
40. $\lim_{x\to 0^+} \left(\dfrac{3x+1}{x}-\dfrac{1}{\sin x}\right)$
41. $\lim_{x\to 1^+} \left(\dfrac{1}{x-1}-\dfrac{1}{\ln x}\right)$
42. $\lim_{x\to 0^+} (\csc x-\cot x+\cos x)$
43. $\lim_{\theta\to 0} \dfrac{\cos\theta-1}{e^\theta-\theta-1}$
44. $\lim_{h\to 0} \dfrac{e^h-(1+h)}{h^2}$
45. $\lim_{t\to \infty} \dfrac{e^t+t^2}{e^t-t}$
46. $\lim_{x\to \infty} x^2e^{-x}$
47. $\lim_{x\to 0} \dfrac{x-\sin x}{x\tan x}$
48. $\lim_{x\to 0} \dfrac{(e^x-1)^2}{x\sin x}$
49. $\lim_{\theta\to 0} \dfrac{\theta-\sin\theta\cos\theta}{\tan\theta-\theta}$
50. $\lim_{x\to 0} \dfrac{\sin 3x-3x+x^2}{\sin x\sin 2x}$

Indeterminate Powers and Products

Find the limits in Exercises 51–66.

51. $\lim_{x\to 1^+} x^{1/(1-x)}$
52. $\lim_{x\to 1^+} x^{1/(x-1)}$
53. $\lim_{x\to \infty} (\ln x)^{1/x}$
54. $\lim_{x\to e^+} (\ln x)^{1/(x-e)}$
55. $\lim_{x\to 0^+} x^{-1/\ln x}$
56. $\lim_{x\to \infty} x^{1/\ln x}$
57. $\lim_{x\to \infty} (1+2x)^{1/(2\ln x)}$
58. $\lim_{x\to 0} (e^x+x)^{1/x}$
59. $\lim_{x\to 0^+} x^x$
60. $\lim_{x\to 0^+} \left(1+\dfrac{1}{x}\right)^x$
61. $\lim_{x\to \infty} \left(\dfrac{x+2}{x-1}\right)^x$
62. $\lim_{x\to \infty} \left(\dfrac{x^2+1}{x+2}\right)^{1/x}$
63. $\lim_{x\to 0^+} x^2\ln x$
64. $\lim_{x\to 0^+} x(\ln x)^2$
65. $\lim_{x\to 0^+} x\tan\left(\dfrac{\pi}{2}-x\right)$
66. $\lim_{x\to 0^+} \sin x\cdot\ln x$

Theory and Applications

L'Hôpital's Rule does not help with the limits in Exercises 67–74. Try it—you just keep on cycling. Find the limits some other way.

67. $\lim_{x\to \infty} \dfrac{\sqrt{9x+1}}{\sqrt{x+1}}$
68. $\lim_{x\to 0^+} \dfrac{\sqrt{x}}{\sqrt{\sin x}}$
69. $\lim_{x\to (\pi/2)^-} \dfrac{\sec x}{\tan x}$
70. $\lim_{x\to 0^+} \dfrac{\cot x}{\csc x}$
71. $\lim_{x\to \infty} \dfrac{2^x-3^x}{3^x+4^x}$
72. $\lim_{x\to -\infty} \dfrac{2^x+4^x}{5^x-2^x}$
73. $\lim_{x\to \infty} \dfrac{e^{x^2}}{xe^x}$
74. $\lim_{x\to 0^+} \dfrac{x}{e^{-1/x}}$

75. Which one is correct, and which one is wrong? Give reasons for your answers.

    a. $\lim_{x\to 3} \dfrac{x-3}{x^2-3} = \lim_{x\to 3} \dfrac{1}{2x} = \dfrac{1}{6}$ **b.** $\lim_{x\to 3} \dfrac{x-3}{x^2-3} = \dfrac{0}{6} = 0$

76. Which one is correct, and which one is wrong? Give reasons for your answers.

    a. $\lim_{x\to 0} \dfrac{x^2-2x}{x^2-\sin x} = \lim_{x\to 0} \dfrac{2x-2}{2x-\cos x}$
    $= \lim_{x\to 0} \dfrac{2}{2+\sin x} = \dfrac{2}{2+0} = 1$

    b. $\lim_{x\to 0} \dfrac{x^2-2x}{x^2-\sin x} = \lim_{x\to 0} \dfrac{2x-2}{2x-\cos x} = \dfrac{-2}{0-1} = 2$

77. Only one of these calculations is correct. Which one? Why are the others wrong? Give reasons for your answers.

a. $\lim_{x\to 0^+} x \ln x = 0 \cdot (-\infty) = 0$

b. $\lim_{x\to 0^+} x \ln x = 0 \cdot (-\infty) = -\infty$

c. $\lim_{x\to 0^+} x \ln x = \lim_{x\to 0^+} \dfrac{\ln x}{(1/x)} = \dfrac{-\infty}{\infty} = -1$

d. $\lim_{x\to 0^+} x \ln x = \lim_{x\to 0^+} \dfrac{\ln x}{(1/x)}$

$$= \lim_{x\to 0^+} \frac{(1/x)}{(-1/x^2)} = \lim_{x\to 0^+} (-x) = 0$$

78. Find all values of c that satisfy the conclusion of Cauchy's Mean Value Theorem for the given functions and interval.

a. $f(x) = x, \quad g(x) = x^2, \quad (a, b) = (-2, 0)$

b. $f(x) = x, \quad g(x) = x^2, \quad (a, b)$ arbitrary

c. $f(x) = x^3/3 - 4x, \quad g(x) = x^2, \quad (a, b) = (0, 3)$

79. Continuous extension Find a value of c that makes the function

$$f(x) = \begin{cases} \dfrac{9x - 3\sin 3x}{5x^3}, & x \neq 0 \\ c, & x = 0 \end{cases}$$

continuous at $x = 0$. Explain why your value of c works.

80. For what values of a and b is

$$\lim_{x\to 0}\left(\frac{\tan 2x}{x^3} + \frac{a}{x^2} + \frac{\sin bx}{x}\right) = 0?$$

T **81. $\infty - \infty$ Form**

a. Estimate the value of

$$\lim_{x\to\infty} \left(x - \sqrt{x^2 + x}\right)$$

by graphing $f(x) = x - \sqrt{x^2 + x}$ over a suitably large interval of x-values.

b. Now confirm your estimate by finding the limit with l'Hôpital's Rule. As the first step, multiply $f(x)$ by the fraction $\left(x + \sqrt{x^2 + x}\right)/\left(x + \sqrt{x^2 + x}\right)$ and simplify the new numerator.

82. Find $\lim_{x\to\infty} \left(\sqrt{x^2 + 1} - \sqrt{x}\right)$.

T **83. 0/0 Form** Estimate the value of

$$\lim_{x\to 1} \frac{2x^2 - (3x + 1)\sqrt{x} + 2}{x - 1}$$

by graphing. Then confirm your estimate with l'Hôpital's Rule.

84. This exercise explores the difference between the limit

$$\lim_{x\to\infty}\left(1 + \frac{1}{x^2}\right)^x$$

and the limit

$$\lim_{x\to\infty}\left(1 + \frac{1}{x}\right)^x = e.$$

a. Use l'Hôpital's Rule to show that

$$\lim_{x\to\infty}\left(1 + \frac{1}{x}\right)^x = e.$$

T **b.** Graph

$$f(x) = \left(1 + \frac{1}{x^2}\right)^x \quad \text{and} \quad g(x) = \left(1 + \frac{1}{x}\right)^x$$

together for $x \geq 0$. How does the behavior of f compare with that of g? Estimate the value of $\lim_{x\to\infty} f(x)$.

c. Confirm your estimate of $\lim_{x\to\infty} f(x)$ by calculating it with l'Hôpital's Rule.

85. Show that

$$\lim_{k\to\infty}\left(1 + \frac{r}{k}\right)^k = e^r.$$

86. Given that $x > 0$, find the maximum value, if any, of

a. $x^{1/x}$ **b.** x^{1/x^2} **c.** x^{1/x^n} (n a positive integer)

d. Show that $\lim_{x\to\infty} x^{1/x^n} = 1$ for every positive integer n.

87. Use limits to find horizontal asymptotes for each function.

a. $y = x \tan\left(\dfrac{1}{x}\right)$ **b.** $y = \dfrac{3x + e^{2x}}{2x + e^{3x}}$

88. Find $f'(0)$ for $f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$

T **89. The continuous extension of $(\sin x)^x$ to $[0, \pi]$**

a. Graph $f(x) = (\sin x)^x$ on the interval $0 \leq x \leq \pi$. What value would you assign to f to make it continuous at $x = 0$?

b. Verify your conclusion in part (a) by finding $\lim_{x\to 0^+} f(x)$ with l'Hôpital's Rule.

c. Returning to the graph, estimate the maximum value of f on $[0, \pi]$. About where is max f taken on?

d. Sharpen your estimate in part (c) by graphing f' in the same window to see where its graph crosses the x-axis. To simplify your work, you might want to delete the exponential factor from the expression for f' and graph just the factor that has a zero.

T **90. The function $(\sin x)^{\tan x}$** (*Continuation of Exercise 89.*)

a. Graph $f(x) = (\sin x)^{\tan x}$ on the interval $-7 \leq x \leq 7$. How do you account for the gaps in the graph? How wide are the gaps?

b. Now graph f on the interval $0 \leq x \leq \pi$. The function is not defined at $x = \pi/2$, but the graph has no break at this point. What is going on? What value does the graph appear to give for f at $x = \pi/2$? (*Hint:* Use l'Hôpital's Rule to find $\lim f$ as $x \to (\pi/2)^-$ and $x \to (\pi/2)^+$.)

c. Continuing with the graphs in part (b), find max f and min f as accurately as you can and estimate the values of x at which they are taken on.

7.6 Inverse Trigonometric Functions

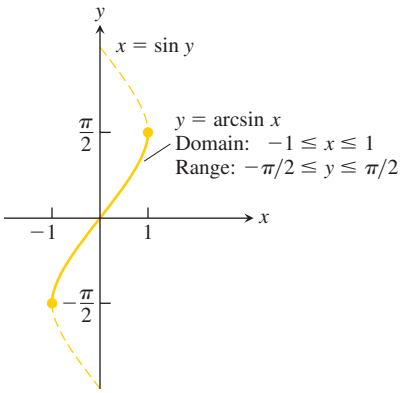


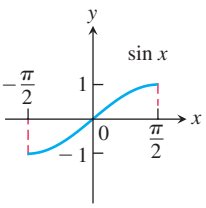
FIGURE 7.21 The graph of $y = \arcsin x$.

Inverse trigonometric functions arise when we want to calculate angles from side measurements in triangles. They also provide useful antiderivatives and appear frequently in the solutions of differential equations. This section shows how these functions are defined, graphed, and evaluated, how their derivatives are computed, and why they appear as important antiderivatives.

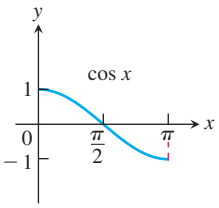
Defining the Inverse Trigonometric Functions

The six basic trigonometric functions are not one-to-one (since their values repeat periodically). However, we can restrict their domains to intervals on which they are one-to-one. The sine function increases from -1 at $x = -\pi/2$ to $+1$ at $x = \pi/2$. By restricting its domain to the interval $[-\pi/2, \pi/2]$ we make it one-to-one, so that it has an inverse which is called arcsin x (Figure 7.21). Similar domain restrictions can be applied to all six trigonometric functions.

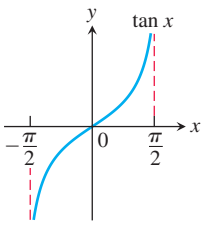
Domain restrictions that make the trigonometric functions one-to-one



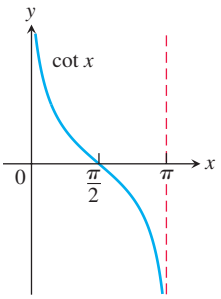
$y = \sin x$
Domain: $[-\pi/2, \pi/2]$
Range: $[-1, 1]$



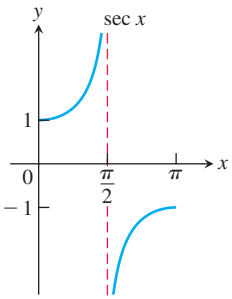
$y = \cos x$
Domain: $[0, \pi]$
Range: $[-1, 1]$



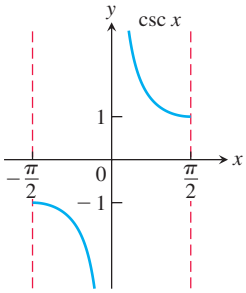
$y = \tan x$
Domain: $(-\pi/2, \pi/2)$
Range: $(-\infty, \infty)$



$y = \cot x$
Domain: $(0, \pi)$
Range: $(-\infty, \infty)$



$y = \sec x$
Domain: $[0, \pi/2) \cup (\pi/2, \pi]$
Range: $(-\infty, -1] \cup [1, \infty)$



$y = \csc x$
Domain: $[-\pi/2, 0) \cup (0, \pi/2]$
Range: $(-\infty, -1] \cup [1, \infty)$

Since these restricted functions are now one-to-one, they have inverses, which we denote by

$$y = \sin^{-1} x \quad \text{or} \quad y = \arcsin x, \qquad y = \cos^{-1} x \quad \text{or} \quad y = \arccos x$$
$$y = \tan^{-1} x \quad \text{or} \quad y = \arctan x, \qquad y = \cot^{-1} x \quad \text{or} \quad y = \operatorname{arccot} x$$
$$y = \sec^{-1} x \quad \text{or} \quad y = \operatorname{arcsec} x, \qquad y = \csc^{-1} x \quad \text{or} \quad y = \operatorname{arccsc} x$$

These equations are read "y equals the arcsine of x" or "y equals arcsin x" and so on.

Caution The -1 in the expressions for the inverse means "inverse." It does *not* mean reciprocal. For example, the *reciprocal* of $\sin x$ is $(\sin x)^{-1} = 1/\sin x = \csc x$. ●